

Problem

For a load, $\mathbf{V}_{\text{rms}} = 110\angle 85^\circ \text{ V}$, $\mathbf{I}_{\text{rms}} = 3\angle 15^\circ \text{ A}$. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Step-by-step solution

Step 1 of 5

(a)

Calculate the value of the complex power $\mathbf{S}$.

$$\mathbf{S} = \mathbf{V}_{\text{rms}} \mathbf{I}_{\text{rms}}^*$$

Substitute $110\angle 85^\circ \text{ V}$ for $\mathbf{V}_{\text{rms}}$ and $3\angle 15^\circ \text{ A}$ for $\mathbf{I}_{\text{rms}}$.

$$\begin{aligned}\mathbf{S} &= (110\angle 85^\circ)(3\angle 15^\circ)^* \\ &= (110\angle 85^\circ)(3\angle -15^\circ) \\ &= 330\angle 85^\circ - 15^\circ \\ &= 330\angle 70^\circ \text{ VA}\end{aligned}$$

Therefore, the value of the complex power $\mathbf{S}$ is, $\boxed{330\angle 70^\circ \text{ VA}}$.

Step 2 of 5

Calculate the value of the apparent power S .

$$S = |\mathbf{V}_{\text{rms}}| |\mathbf{I}_{\text{rms}}|$$

Substitute 110 V for $|\mathbf{V}_{\text{rms}}|$ and 3 A for $|\mathbf{I}_{\text{rms}}|$.

$$S = 330 \text{ VA}$$

Therefore, the value of the apparent power S is, $\boxed{330 \text{ VA}}$.

Step 3 of 5

(b)

Consider the value of the complex power $\mathbf{S}$.

$$\mathbf{S} = 330\angle 70^\circ \text{ VA}$$

Convert to rectangular coordinate system.

$$\begin{aligned}\mathbf{S} &= 330(\cos 70^\circ + j \sin 70^\circ) \\ &= 330(0.34202 + j0.93969) \\ &= 112.87 + j310.1 \text{ VA}\end{aligned}$$

Compare $112.87 + j310.1 \text{ VA}$ with $P + jQ$, where P represents real power and Q represents reactive power.

Therefore, the value of the real power P is $\boxed{112.87 \text{ W}}$ and the value of the reactive power is Q is, $\boxed{310.1 \text{ VAR}}$.

Step 4 of 5

(c)

Calculate the value of the power factor pf .

$$\text{pf} = \cos(\theta_v - \theta_i)$$

Substitute 85° for θ_v and 15° for θ_i .

$$\begin{aligned}\text{pf} &= \cos(85^\circ - 15^\circ) \\ &= \cos(70^\circ) \\ &= 0.342 \text{ lagging}\end{aligned}$$

The power factor is lagging since the value of the reactive power is positive.

Therefore, the value of the power factor pf is, $\boxed{0.342}$.**Step 5** of 5Calculate the value of the load impedance Z .

$$Z = \frac{V_{\text{rms}}}{I_{\text{rms}}}$$

Substitute $110\angle 85^\circ \text{ V}$ for V_{rms} and $3\angle 15^\circ \text{ A}$ for I_{rms} .

$$\begin{aligned}Z &= \frac{110\angle 85^\circ}{3\angle 15^\circ} \\ &= 36.667\angle 70^\circ \\ &= 12.541 + j34.46 \Omega\end{aligned}$$

Therefore, the value of the load impedance Z is, $\boxed{12.541 + j34.46 \Omega}$.